Contact

mike.bobak+in@gmail.com

www.linkedin.com/in/michaelbobak (LinkedIn)
mbcode.github.io/ (Personal)
represent2reason-inc.github.io/ (Company)

Top Skills

metaphactory
Knowledge Acquisition
Linked Data

Languages

German (Elementary)
English (Native or Bilingual)

Certifications

Semantic Web Technologies
Information Service Engineering
Knowledge Engineering with Semantic Web Technologies
Knowledge Graphs
Knowledge Graphs - Foundations and Applications

Publications

A turbulence model for sound propagation from an elevated source above level ground

Mike Bobak

Sr Semantic Web Engineer at IMO Health
Champaign, Illinois, United States

Summary

Creative computational problem-solving done as a: Ontology/ Knowledge-Engineer, Research-Programmer/Analyst/Architect, Systems-Analyst, Multi-disciplined (Systems) Engineer, or Software-Engineer.

I have a strong desire to further my knowledge and experience with a variety of Artificial Intelligence and simulation techniques. I hope to do this by peer interaction on a challenging project. I am particularly interested in projects with any part/aspect of Knowledge-Based-Simulation Environment/(Semantic Digital Twin) &/or something like an Assisted Problem-Solving-Environment. Driving-force/ Problem: Many computational problems would be best solved by a set of applications/knowledge-sources that usually can not work together well manually, much less in an automated way. Model-Based-Reasoning/Composable-Simlation as it is done now takes too long and does not make use of pre-built software. I would like to continue to get past these problems. The domain of greatest interest, is any science with a computational component. Where there is some model of what is going on, even if not that explicit. I want to take simulation models,(live)experimental data/whatever is available, and allow them all to contribute to a solution. I would particularly like to work on research in this area. I am willing to help write any related grant applications, and take on a bulk of the work. As my last 10-15+yrs of work in various related work has uniquely prepared me, and I would like to get started asap.

Specialties: Knowledge-Engineering (Kn/Rule-Based, Kn-Rep&Reasoning) Sci-apps if possible. More at: https:// mbcode.github.io/

Experience

IMO Health

Sr Semantic Web Engineer
August 2026 - Present (3 months)

Represent2Reason Inc ->PanaceaDirect->CitiusTech-> IMO_Health

Similar to the vertex work, but more healthcare focused and at a larger scale

Pivot Path Solutions
Subject Matter Expert
April 2026 - August 2026 (5 months)

SME for digital-twin efforts, which incl. the use of semantics

First for a DARPA medical digital-twin but then for a general path forward

Vertex Pharmaceuticals
Ontology Data Analyst (Integrations)
December 2023 - December 2025 (2 years 1 month)
remote

Contract through Xendat Data & Analytics to the pharma's Ontology Group

Focusing on automation to scale ontology construction. Use of Python and semantic tools: SPARQL, Triple-Stores, (web)Protege, Semantic-Web-Libs

Wrote new libs used for both a Semantic-ETL and for an automated QA/QC, aiding Ontologist & BizDev, incl several apps

Aloha Health Network
Sr Knowledge Engineer
June 2018 - April 2025 (6 years 11 months)
remote

- Part of a startup built on the topic of my UCSF research. Semantic search for clinical trials. [graph store/s]

- Intelligent document processing, of pharma CofAs etc, automation that also allows for better analytics

part time startup, that finally shut down

National Center for Supercomputing Applications
Sr Research Software Engineer
December 2019 - August 2023 (3 years 9 months)
Urbana-Champaign Area

- All the PoCs for earthcube.org/geocodes incl. organizing others to bring in a great NSF review

- Focus on semantics/metadata search, with some NLP and sim. [Python/ SPARQL ...]

Nutrien Ag Solutions

Sr Software Engineer Natural Systems

June 2018 - June 2019 (1 year 1 month)

Urbana-Champaign Area

Planned & guided reworking the main simulation, it's documentation, & ML/ verification

Helped get Agrible in a position to be acquired by Nutrien

consulting

freelance/ side project/s

July 2011 - June 2018 (7 years)

San Francisco Bay Area

Freelance, develop startup idea/s, which started with working on a ProofOfConcept for Patient Data-Mining-Cluster patent application that I helped start at ucsf.edu, work on an assisted eco-sim/modeling environment in Lisp, and semantic-web (industrial) IoT advice...

Apollo Group

Architect Adaptive-Learning-Platform

October 2010 - July 2011 (10 months)

San Francisco Bay Area

Conceptually annotate study material & tests for automated remediation, instrument classroom to learn from use Lisp/KM Hadoop

UCSF

Programmer/Analyst III

2007 - October 2010 (3 years)

San Francisco Bay Area

Medical-Informatics research (relating to clinical-trails), and Natural-Language-Processing in Java/etc; paper with Stanford group; ontology dev/ use http://rctbank.ucsf.edu/

https://web.archive.org/web/20100808164110/

https://bmir.stanford.edu/publications/view.php/a_practical_method_for_transforming_free_text_eligibility_criteria_into_computable_c

knowledge-based-consulting

Knowledge Engineer

February 2001 - September 2007 (6 years 8 months)

Greater Chicago Area

2/2001-present Freelance/Mindbox Chicago, IL

Knowledge Engineer

- Rule-based, Case-based, Machine-Learning/Data-Mining, &/or general (Lisp) programming.
- Long-term computational-science (bio/chem/physics,bioinformatics), sim/AI background.
- Research(science/AI/etc)programming, simulation, visualization,...
- Worked through mindbox.com 3/02-10/02. [used Art*Enterprise]
See: "http://softdist.mindbox.com/pressreleases/Ocwen MindBox.doc"
- Worked up to 1/2time max for cas.dis.anl.gov 5/03-5/04 [Java Simulation]
see: http://www.swarm.org/pipermail/support/2005-February/014507.html
- Worked fulltime 8/03-~05 labs.gte.com, Model-Based-Diagnosis application used on a national scale. On leave for contracts now. [used Art*Enterprise]
See: http://mike.bobak.googlepages.com/IAAI96-SSCFI.pdf
- Bioinformatics/control contract 11/04-present [using CLIPS&Protege]
GHX.com med-Lisp, and now ucsf.edu fulltime
more at: http://mike.bobak.googlepages.com/

Verizon
programmer
2003 - 2005 (2 years)
waltham ma

Model-Based-Diagnosis on a n
ational scale. [Art *Enterprise] See: aaai.org/Papers/IAAI/1996/IAAI96-287.pdf

anl.gov
contractor
2003 - 2003 (less than a year)

Mindbox
Knowledge Engineer
2002 - 2002 (less than a year)

http://www.mindbox.com/NewsEvents/PressReleases/21OCT2002.pdf

University of Illinois at Urbana-Champaign
(Senior) Research Programmer
1998 - 2001 (3 years)
Urbana-Champaign Area

Organize many levels of a very large knowledge based simulation projects. Brought over 18 programmers together to deliver a coherent product. Ran weekly (sub)group meetings, down to help solving any problem. Hiring, demo/design/install trips, prototyping to lead project direction. Taught group of 6 how

to use a Rule-Based-shell for a reasoner-rewrite in Art*Enterprise. Projects included: Simulation-based, Intelligent Tutoring System (ITS) & Real-Time control system. Being used in classroom, real life testing, presented at IAAI99 'Automated Instructor Assistant for Ship Damage Control' The system teaches Navy officers how to save a simulated ship in crisis. A variant was developed to catch real crisis conditions and suggest solutions, in real-time.

brightware
Knowledge Engineer
October 1996 - June 1998 (1 year 9 months)
Greater Chicago Area

Helped develop and install their very first product (Intelligent email reply). Worked between development and consulting. Helped on several Knowledge-Based business applications. Helped with several deployed Knowledge-Based business applications (ie. financial: mortgage, web based job finder). All with Art*Enterprise. See: http://www.brightware.com/eservice_solutions/ More recently I worked 1/2year for the new version of the company: Mindbox.

Institute of Learning Sciences Northwestern.edu
Lead Programmer/Analyst
February 1996 - August 1996 (7 months)
Evanston IL

Wrote Lisp code (mainly GUI) for Qualitative Research Group. Learned more about Qualitative/Quantitative Simulation, Model-Based Reasoning, Intelligent-Tutoring- Systems, & general Lisp programming. See: http://www.qrg.northwestern.edu/projects/NSF/Cyclepad/aboutcp.htm

Argonne National Laboratory
Software Engineer
February 1993 - February 1996 (3 years 1 month)
Lemont IL

(EAD then DIS groups) Wrote fielded Expert System by myself at the end of grad-school. [in Lisp rule-shell then CLIPS] Prototyped communication & control of distributed simulation. [in CLIPS PVM etc] Agent wrapping of simulations with CLIPS+PVM, to describe then mix and match them. Also used C++/Smalltalk/FORTRAN with PVM; Other work as needed. Algo/Viz/Etc. Written up in a book about innovative distributed object application. See: http://www.dis.anl.gov/DEEM http://www.dis.anl.gov/DEEM/DIAS http://mike.bobak.googlepages.com/bobak/diaswp.pdf _More recently I worked part-time for the new subgroup of dis: cas.dis.anl.gov.

uiuc.edu
Graduate Research Assistant /Research Programmer
January 1990 - January 1993 (3 years 1 month)
Urbana-Champaign Area

Wrote molecular graphics package used in classes & for publications. [in C] Used machine-learning techniques for protein structure prediction. Wrote thesis on Knowledge-Based Simulation Environment. [Lisp/OPS5/C] Overseen by heads of the NCSA CompBio group and head of Biophysics at the time. see: http://web.bilkent.edu.tr/ncsa/Apps/CBdir.html

NCSA UIUC &GIST
Programmer/Consultant
April 1989 - December 1989 (9 months)
Urbana-Champaign Area

Suggested scientific software path for Software Tools Group of NCSA; Wrote molecular viz code for a professor. Wrote testing code for Global Info Systems Tech. [in C]

Shearson Lehman Hutton
Programmer (Research Computing)
October 1988 - April 1989 (7 months)

Maintained financial databases & daily report information. Organized worldwide mailing system. Wrote statistics code for stock predictions. [MUMPS and Maths-package]

US Corp of Engineers Construction Engineering Research Lab
Research Programmer (Modeling then Acoustics teams)
March 1982 - August 1988 (6 years 6 months)
Urbana-Champaign Area

Provided research support from start to finish. Wrote and ran computer simulation code, compared output with field data. [FORTRAN] Did field measurments to back up predictions. (Team/Self; Local/US/World-wide) My work went into several published papers. See:http://adsabs.harvard.edu/cgi-bin/nph-bib_query?1987ASAJ...81..638J & 1987nce..conf..215R http://www.cecer.army.mil/td/tips/pub/details.cfm?PUBID=1452&TOP=1 Early work went into GRASS: http://grass.itc.it/intro/general.php

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Education

University of Illinois Urbana-Champaign

MS, Computational Sciences /AI · (1990 - 1993)

University of Illinois Urbana-Champaign

BS, Physics and Biophysics · (1983 - 1988)